

HONOLULU INTL, HI, USA

WMO: 911820

Lat: 21.324N

Lon: 157.929W

Elev: 7

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22521

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.5	65.1	51.5	56.6	71.3	53.6	61.0	71.3	24.7	76.0	22.1	75.9	4.2	320	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	89.6	74.1	88.7	73.7	87.8	73.3	77.4	84.5	76.4	83.9	75.6	83.3	15.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	132.7	81.2	74.2	127.6	80.7	73.1	123.1	80.2	40.7	84.6	39.7	83.7	38.9	83.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.6	20.5	19.0	DB	59.4	90.8	2.2	1.8	57.9	92.1	56.6	93.2	55.4	94.2	53.8	95.5	
			WB	55.5	78.5	1.9	1.6	54.1	79.7	53.0	80.6	51.9	81.5	50.5	82.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	73.7	73.9	74.6	76.5	78.0	80.1	81.4	82.0	81.5	80.2	77.8	75.3
	DBStd	3.64	2.46	2.41	2.33	2.27	2.31	1.69	1.69	1.63	1.81	2.00	2.25	2.38
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10196	735	669	763	794	867	904	972	991	944	938	834	785
	CDD65	4721	270	249	298	344	402	454	507	526	494	473	384	320
	CDH74	36442	1207	1075	1397	1989	2959	3998	4942	5372	4874	4277	2700	1652
	CDH80	8930	75	67	131	328	652	1106	1508	1740	1538	1143	485	157
Wind	WSAvg	10.3	8.4	9.1	9.9	11.0	10.3	11.7	12.0	11.8	10.3	9.8	9.7	9.3
Precipitation	PrecAvg	18.8	2.7	2.0	2.0	1.1	1.0	0.4	0.6	0.6	0.8	1.8	2.4	3.0
	PrecMax	42.8	12.8	9.4	8.4	8.9	7.2	2.4	2.7	7.6	4.5	11.1	14.7	17.3
	PrecMin	3.7	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0
	PrecStd	8.6	2.9	2.2	1.9	1.5	1.3	0.6	0.6	1.1	0.8	2.0	2.9	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.5	83.7	84.3	86.5	88.1	89.4	90.0	91.1	90.8	89.4	87.4	85.4
		MCWB	70.8	71.3	69.6	71.7	72.5	72.6	74.5	74.7	75.0	74.1	73.8	73.6
	2%	DB	81.9	81.9	82.8	84.7	86.3	87.7	88.8	89.6	89.2	88.1	85.6	83.0
		MCWB	70.0	70.0	69.6	70.6	72.0	72.5	73.6	74.1	74.5	73.8	72.7	70.9
	5%	DB	80.6	80.6	81.5	83.4	85.0	86.5	87.7	88.4	88.1	86.7	84.2	81.7
		MCWB	69.6	69.4	69.6	70.1	71.3	72.1	73.1	73.5	73.9	73.3	72.1	70.0
	10%	DB	79.4	79.4	80.2	82.0	83.6	85.4	86.6	87.3	86.9	85.4	82.9	80.4
		MCWB	69.2	69.0	69.2	69.8	70.8	71.6	72.7	72.9	73.4	72.7	71.6	69.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.1	74.9	74.2	75.1	76.5	76.5	77.4	78.2	78.9	77.7	77.0	76.0
		MCDB	79.5	80.0	79.8	81.8	82.9	84.0	84.9	85.1	86.1	84.5	82.5	81.5
	2%	WB	73.5	73.0	72.8	73.2	74.5	75.0	76.1	76.7	77.5	76.3	75.4	73.9
		MCDB	77.9	78.6	78.7	80.7	82.2	83.2	84.3	84.8	84.8	83.5	81.3	79.2
	5%	WB	72.2	71.7	71.7	72.1	73.2	73.7	75.1	75.5	76.3	75.3	74.4	72.7
		MCDB	77.5	78.0	77.8	79.8	81.9	83.0	83.8	84.4	84.1	82.8	80.8	78.4
	10%	WB	71.1	70.6	70.7	71.1	72.2	72.9	74.1	74.7	75.3	74.5	73.3	71.5
		MCDB	77.1	77.1	77.3	79.0	81.0	82.4	83.2	83.8	83.4	82.3	80.2	77.9
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.7	11.6	11.5	11.8	11.6	11.5	11.8	12.0	11.4	10.8	11.1
		MCDBR	12.8	12.3	12.6	12.4	12.6	12.3	12.0	12.6	12.7	12.2	11.6	11.7
		MCWBR	5.8	5.4	5.2	4.4	4.5	4.1	3.9	3.9	4.1	4.2	4.5	5.0
	5% WB	MCDBR	11.1	11.3	11.4	11.7	11.8	11.5	11.5	11.9	11.8	11.3	10.6	10.6
	MCWBR	6.3	6.1	6.0	5.1	5.3	4.8	4.5	4.5	4.8	4.9	5.3	5.8	
Clear-Sky Solar Irradiance	taub	0.363	0.370	0.379	0.391	0.384	0.371	0.370	0.368	0.362	0.377	0.376	0.359	
	taud	2.575	2.544	2.530	2.481	2.532	2.575	2.590	2.593	2.615	2.576	2.571	2.610	
	Ebn at Noon	289	293	294	290	288	290	290	292	295	287	283	287	
	Edh at Noon	29	32	34	36	34	32	31	31	30	30	29	28	
All-Sky Solar Radiation	RadAvg	1410	1587	1818	2052	2196	2297	2261	2173	2031	1708	1441	1303	
	RadStd	61	112	191	117	125	65	47	86	61	54	38	94	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+122

Nomenclature: See separate page